# DeepStream license plate detection and recognition (LPR)

**DeepStream license plate detection and recognition (LPR)**

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> Applicable environment: JetPack 7.2.1, Ubuntu 24.04, DeepStream 9.1, Jetson Orin NX 16GB.

> The factory environment already has the China and the US license plate models, the TensorRT engines, the LPR program and the course entry. In normal use you start directly from section 4; redeployment is only needed when the program or the models are missing.

## 1. NVIDIA official deployment (for reference only)

**The current factory system has already completed the deployment described above, so you do not need to repeat the following operations.**

Official source code and documentation: https://github.com/NVIDIA/DeepStream/tree/main/src/apps/tao_apps/apps/tao_others/deepstream_lpr_app

The official LPR sample is located in the DeepStream source tree. Only when the program, the models or the engines are missing and the environment has to be restored, execute the following steps in this order:

```
# Download the TAO models; the first download requires network access
cd ~/deepstream_tao_apps
~/deepstream_tao_apps/download_models.sh

# Build the LPR application and the custom text parser
cd ~/deepstream_tao_apps/apps/tao_others/deepstream_lpr_app
make
```

The model download location:

```
~/deepstream_tao_apps/models
```

On the first actual run, DeepStream automatically generates the TensorRT engine according to the GPU, CUDA and TensorRT versions of the local machine. The official manual run method:

```
cd ~/deepstream_tao_apps/apps/tao_others/deepstream_lpr_app

# China version
cp dict_ch.txt dict.txt
./deepstream-lpr-app ../../../configs/app/lpr_app_ch_config.yml

# US version
cp dict_us.txt dict.txt
./deepstream-lpr-app ../../../configs/app/lpr_app_us_config.yml
```

`download_models.sh` is located in the root directory `~/deepstream_tao_apps/`, not in the `deepstream_lpr_app` subdirectory. At the end the script may try to download an additional model into the old Tracker path; that model does not belong to the three-stage LPR chain of vehicle, license plate and text recognition.

# 2. Case goal and processing chain

This case recognizes the license plate on a vehicle and displays the result in real time on the Jetson device itself or on the VNC desktop.

The current implementation uses the three-stage chain of the official NVIDIA TAO LPR:

```
TrafficCamNet vehicle detection
        ↓
LPD license plate location detection
        ↓
LPR license plate character recognition
```

Therefore the LPD only looks for the license plate inside the regions of the vehicles that have already been detected. When a license plate is filmed on its own, detached from a vehicle, the preceding vehicle detection produces no result, and the subsequent license plate detection and text recognition are not executed.

# 3. Model version and applicable scope

| Parameter | Model used | Suitable license plates |
|---|---|---|
| `--cn` | `LPD_ch` + `LPR_ch` | Mainland China license plates, including the province abbreviation, such as `粤A83288` and `皖A52718` |
| `--us` | `LPD_us` + `LPR_us` | US state license plates |

The China version keeps the UTF-8 Chinese province abbreviation; the stabilization processing must not reduce it to only the trailing letters and digits.

The US version does not suit long European plates such as the Romanian one. Adjusting the threshold can improve the recall and the stability, but it cannot compensate for the fundamental differences in the model's region, the plate proportion and the character layout.

# 4. Course script entry

The current unified startup script is located in `~/deepstream_tao_demo/`:

```
cd ~/deepstream_tao_demo
```

| Input source | China version | US version |
| --- | --- | --- |
| Local video | `./run_lpr_cn.sh` | `./run_lpr_us.sh` |
| USB camera | `./run_lpr_cn_usb.sh` | `./run_lpr_us_usb.sh` |
| CSI camera | `./run_lpr_cn_csi.sh` | `./run_lpr_us_csi.sh` |

All three entries:

1. Switch to the corresponding character dictionary: `dict_ch.txt` or `dict_us.txt`.
2. Use the corresponding China or US model configuration.
3. Set the DeepStream/CUDA runtime library path.
4. Display the real-time window on the desktop of the Jetson device itself or on the VNC desktop.

Before running USB/CSI, confirm that no other program is occupying the corresponding camera.

Output rule: the local video entry displays in real time and saves an MP4 at the same time; the USB and CSI entries only display in real time and do not generate a recording file.

# 5. Run with a local video

Run:

```
cd ~/deepstream_tao_demo && export DISPLAY=:0 && ./run_lpr_cn.sh
# US version: cd ~/deepstream_tao_demo && export DISPLAY=:0 && ./run_lpr_us.sh
```




The video location of the current configuration:

| Version | Configuration file | Default video |
|---|---|---|
| China version | `~/deepstream_tao_apps/deepstream_app_tao_configs/run_lpr_china.txt` | `file:///home/jetson/Videos/Video007.mp4` |
| US version | `~/deepstream_tao_apps/deepstream_app_tao_configs/run_lpr_us.txt` | Prefer `~/Videos/us.mp4`, otherwise use the bundled `Test cases/us.mp4` of the course |

To replace the video, pass the absolute path directly to `run_lpr_cn.sh` or `run_lpr_us.sh`; the script generates a temporary configuration and does not modify the repository template. The file input of this example supports MP4/H.264 only.

After the video finishes playing naturally, `lpr_cn.mp4` or `lpr_us.mp4` with the recognition boxes and the text is generated. After the script exits, it uses `ffprobe` to verify the container of the file.

To stop, press `Ctrl+C` only once and wait for the application to exit before you check the file; do not interrupt repeatedly and do not use `kill -9`.

# 6. Real-time run with the USB camera

Run:

```
cd ~/deepstream_tao_demo && export DISPLAY=:0 && ./run_lpr_cn_usb.sh
# US version: cd ~/deepstream_tao_demo && export DISPLAY=:0 && ./run_lpr_us_usb.sh
```




## 7. Real-time run with the CSI camera

Run:

```
cd ~/deepstream_tao_demo && export DISPLAY=:0 && ./run_lpr_cn_csi.sh
# US version: cd ~/deepstream_tao_demo && export DISPLAY=:0 && ./run_lpr_us_csi.sh
```



# 8. Stopping and basic troubleshooting

The real-time window is displayed on the Jetson device itself or on the VNC desktop. To stop it, go back to the terminal where the startup command runs and press `Ctrl+C` once.

| Symptom | What to check first |
| --- | --- |
| Only vehicle boxes, no license plate boxes | Whether the license plate is clear, complete and large enough; whether the model version matches the region of the plate. |
| The China plate shows only letters and digits | Make sure that `run_lpr_cn.sh` is used, not `run_lpr_us.sh`. |
| The USB fails to start | Run `v4l2-ctl -d /dev/video0 --list-formats-ext` and confirm that the camera is not occupied by another program. |
| No picture with CSI | Check `sensor-id=0`, the Argus and the camera occupancy; the script directly uses `type=5` and does not wrongly open `/dev/video10`. |